

PhosphoAtlas Agent Benchmark

Summary Tables

Peptide accuracy = case-insensitive (biological identity match)

Table 1. Overall performance comparison across models (naive prompt condition).

Model	Type	Entries	Recall	Prec	F1	Kinases	Pep Acc	Multi-DB	DBs Used
Claude Sonnet 4.6	Proprietary	15,434	0.8727	0.9007	0.8865	404/433	100.0%	0.0%	PhosphoSitePlus
Claude Opus 4.6	Proprietary	20,695	0.9524	0.7254	0.8235	417/433	99.8%	29.9%	PhosphoSitePlus, SIGNOR
Qwen 3.2 35B	Open-source	9,671	0.4629	0.7483	0.5720	377/433	95.6%	0.0%	SIGNOR
GPT-5 (OpenCLAW)	Proprietary	11,142	0.4826	0.6832	0.5656	382/433	0.0%	0.0%	PhosphoSIGNOR
Gemini 3.1 Pro	Proprietary	1,428	0.0500	0.5476	0.0916	61/433	0.0%	8.3%	N/A
Mistral Large	Proprietary	15	0.0008	0.8000	0.0016	9/433	0.0%	33.3%	PhosphoSitePlus, SIGNOR
Gemini 2.5 Flash	Proprietary	9	0.0000	0.0000	0	0/433	0	0.0%	UniProt, UniProtKB
Gemini 3.0 Flash	Proprietary	9	0.0000	0.0000	0	0/433	0	0.0%	UniProt, UniProtKB

Table 2. Effect of prompt engineering on model performance.

Model	Condition	Recall	F1	Pep Acc	Multi-DB	vs Naive
Claude Opus 4.6	naive	0.9524	0.8235	99.8%	29.9%	--
Claude Opus 4.6	paper_informed	0.9524	0.8235	99.8%	29.9%	No change
Claude Opus 4.6	pipeline_guided	0.9524	0.8235	99.8%	29.9%	No change
Gemini 3.1 Pro	naive	0.0500	0.0916	0.0%	8.3%	--
Gemini 3.1 Pro	paper_informed	0.0404	0.0760	0.0%	0.0%	-0.010 R
Gemini 3.1 Pro	pipeline_guided	0.0309	0.0591	0.0%	6.3%	-0.019 R
Mistral Large	naive	0.0008	0.0016	0.0%	33.3%	--
Mistral Large	paper_informed	0.0003	0.0006	0.0%	0.0%	No change
Qwen 3.2 35B	explicit_prompt	0.4730	0.5720	85.5%	0.0%	+0.010 R
Qwen 3.2 35B	naive	0.4629	0.5720	95.6%	0.0%	--
Qwen 3.2 35B	paper_informed	0.4629	0.5720	95.6%	0.0%	No change
Qwen 3.2 35B	pipeline_informed	0.4730	0.5720	85.5%	0.0%	+0.010 R

Table 3. Claude Opus vs Sonnet adjusted comparison (assuming FPs are valid novel entries).

Metric	Opus	Sonnet	Winner
Gold entries found (TP)	14,890	13,644	Opus (+1,246)
Novel entries (likely valid)	5,637	1,505	Opus (+4,132)
Total real data curated	20,527	15,149	Opus (+5,378)
Kinases found	417/433	404/433	Opus
Multi-DB cross-validation	29.9%	0.0%	Opus
Peptide accuracy (case-insensitive)	99.8%	100.0%	Comparable
Peptide true mismatches	34/14101	7/13601	Sonnet
UniProt accuracy	91.9%	99.7%	Comparable

Table 4. Tool call efficiency across models and conditions.

Model	Condition	Tool Calls	Runtime	DBs Queried	Entries
Claude Opus 4.6	naive	8	13s	PhosphoSitePlus, SIGNOR	20,695
Claude Opus 4.6	paper_informed	5	9s	PhosphoSitePlus, SIGNOR	20,695
Claude Opus 4.6	pipeline_guided	5	9s	PhosphoSitePlus, SIGNOR	20,695
Claude Sonnet 4.6	naive	31	10.0m	PhosphoSitePlus	15,434
Gemini 3.1 Pro	naive	?	?	N/A	1,428
Gemini 2.5 Flash	naive	?	?	UniProt, UniProtKB	9
Gemini 3.0 Flash	naive	?	?	UniProt, UniProtKB	9
Gemini 3.1 Pro	paper_informed	?	?	N/A	979
Gemini 3.1 Pro	pipeline_guided	?	?	N/A	718
Mistral Large	naive	3	110s	PhosphoSitePlus, SIGNOR	15
Mistral Large	paper_informed	4	91s	PhosphoSitePlus	5
GPT-5 (OpenCLAW)	naive	?	?	PhosphoSIGNOR	11,142

Table 5. Per-tier recall by kinase size (naive prompt condition).

Model	Tier A (100+)	Tier B (20-99)	Tier C (5-19)	Tier D (<5)
Claude Sonnet 4.6	0.8811	0.8626	0.8698	0.7700
Claude Opus 4.6	0.9612	0.9398	0.9449	0.8914
Qwen 3.2 35B	0.4626	0.4496	0.4924	0.5176
GPT-5 (OpenCLAW)	0.4848	0.4657	0.5048	0.5463
Gemini 3.1 Pro	0.0552	0.0404	0.0530	0.0128
Mistral Large	0.0013	--	--	--
Gemini 2.5 Flash	--	--	--	--
Gemini 3.0 Flash	--	--	--	--

Table 6. Column-level accuracy for matched entries (naive prompt condition). Pep Acc = case-insensitive match rate and

Model	Matched	Pep Acc*	Hallucinated	Missing	Pep Total	UniProt Acc
Claude Sonnet 4.6	13,644	99.9%	7	0	13601	99.7%
Claude Opus 4.6	14,890	99.8%	34	0	14101	91.9%
Qwen 3.2 35B	7,237	95.6%	283	0	6454	98.9%
GPT-5 (OpenCLAW)	7,545	0.0%	0	6760	6760	98.7%
Gemini 3.1 Pro	782	0.0%	521	254	775	96.2%
Mistral Large	12	0.0%	12	0	12	91.7%
Gemini 2.5 Flash	0	0.0%	0	0	0	0
Gemini 3.0 Flash	0	0.0%	0	0	0	0

Table 7. Token usage and computational cost.

Model	Condition	Input Tok	Output Tok	Total Tok	Tool Calls	Est. Cost
Claude Opus 4.6	naive	163,644	3,571	167,215	8	\$2.72
Claude Opus 4.6	paper_informed	80,516	2,293	82,809	5	\$1.38
Claude Opus 4.6	pipeline_guided	81,231	2,293	83,524	5	\$1.39